

Stelios Tsampas

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Current Position

Tenure-track Assistant Professor

Feb. 2025 – present

ACP Section, IMADA at the University of Southern Denmark (SDU)

Past Positions

Postdoctoral Researcher

Sep. 2021 – Jan. 2025

Chair of Theoretical Computer Science at **FAU Erlangen-Nürnberg**

- 09.2021 – 12.2023, CoMoC: *Coalgebraic Model Checking*
- from 03.2024, ATLaS: *Abstract Techniques for Programming Languages and Secure Compilation* (personal project)

Doctoral Researcher

Aug. 2016 – July 2021

DistriNet Research Unit at **KU Leuven**

Education

KU Leuven

2016 – 2021

Ph.D. in Computer Science

- *Thesis*: Formal Reasoning about Equivalences:
from Secure Compilation to Categorical Weak Bisimulation
- *Advisor*: Prof. Frank Piessens

Aristotle University of Thessaloniki

2006 – 2014

Master in Electrical and Computer Engineering

- *Thesis*: A Vulnerable System and Wargame
- *Advisor*: Prof. Andreas L. Symeonidis

Research Interests

Foundations of Programming Languages; Computer Security; Formal Methods; Verified Compilation; Categorical Structures in Computer Science

Grants

Deutsche Forschungsgemeinschaft (DFG):

ATLaS: *Abstract Techniques for Programming Languages and Secure Compilation*

PIs: Stelios Tsampas, Sergey Goncharov. €674.636 (€348.725 personal share). 01.2024 – 12.2026.

Advising

Stefano Volpe, co-advised with Marco Peressotti, from 09.2025, ATLaS,

Pouya Partow, co-advised with Sergey Goncharov, from 03.2024, ATLaS.

Publications (Conference)

- Bialgebraic Reasoning on Stateful Languages** ICFP 2025
Sergey Goncharov, Stefan Milius, Lutz Schröder, Stelios Tsampas, Henning Urbat
ACM SIGPLAN International Conference on Functional Programming
- CRDT Emulation, Simulation, and Representation Independence** ICFP 2025
Nathan Liittschwager, Jonathan Castello, Stelios Tsampas, Lindsey Kuper
ACM SIGPLAN International Conference on Functional Programming
- Big Steps in Higher-Order Mathematical Operational Semantics** ICFP 2025
Sergey Goncharov, Pouya Partow, Stelios Tsampas
ACM SIGPLAN International Conference on Functional Programming
- Abstract Operational Methods for Call-by-Push-Value** POPL 2025
Sergey Goncharov, Stelios Tsampas, Henning Urbat
ACM SIGPLAN Symposium on Principles of Programming Languages
- Bialgebraic Reasoning on Higher-Order Program Equivalence** LICS 2024
Sergey Goncharov, Stefan Milius, Stelios Tsampas, Henning Urbat
ACM/IEEE Symposium on Logic in Computer Science
- Logical Predicates in Higher-Order Mathematical Operational Semantics** FoSSaCS 2024
Sergey Goncharov, Alessio Santamaria, Lutz Schröder, Stelios Tsampas, Henning Urbat
Foundations of Software Science and Computation Structures
- Weak Similarity in Higher-Order Mathematical Operational Semantics** LICS 2023
Henning Urbat, Stelios Tsampas, Sergey Goncharov, Stefan Milius, Lutz Schröder
ACM/IEEE Symposium on Logic in Computer Science
- Towards a Higher-Order Mathematical Operational Semantics** POPL 2023
Sergey Goncharov, Stefan Milius, Lutz Schröder, Stelios Tsampas, Henning Urbat
ACM SIGPLAN Symposium on Principles of Programming Languages
Invited for submission to J. Funct. Program.
- Stateful Structural Operational Semantics** FSCD 2022
Sergey Goncharov, Stefan Milius, Lutz Schröder, Stelios Tsampas, Henning Urbat
International Conference on Formal Structures for Computation and Deduction
- CapablePtrs: Securely Compiling Partial Programs** CSF 2021
Using the Pointers-as-Capabilities Principle
Akram El-Korashy, Stelios Tsampas, Marco Patrignani, Dominique Devriese, Deepak Garg, Frank Piessens
IEEE Computer Security Foundations Symposium
- Abstract Congruence Criteria for Weak Bisimilarity** MFCS 2021
Stelios Tsampas, Christian Williams, Andreas Nuyts, Dominique Devriese, Frank Piessens
International Symposium on Mathematical Foundations of Computer Science
- Fully Abstract and Robust Compilation, and How to Reconcile the Two, Abstractly** APLAS 2021
Carmine Abate, Matteo Busi, Stelios Tsampas
Asian Symposium on Programming Languages and Systems
- A Categorical Approach to Secure Compilation** CMCS 2020
Stelios Tsampas, Andreas Nuyts, Dominique Devriese, Frank Piessens
Coalgebraic Methods in Computer Science

Temporal Safety for Stack Allocated Memory on Capability Machines

CSF 2019

Stelios Tsampas, Dominique Devriese, Frank Piessens

IEEE Computer Security Foundations Symposium

Manuscripts Under Review (Journal and Conference)

Higher-Order Mathematical Operational Semantics

Sergey Goncharov, Stefan Milius, Lutz Schröder, Stelios Tsampas, Henning Urbat

Journal of Functional Programming

Other Publications

CRDTs, Coalgebraically (Early Ideas)

CALCO 2023

Nathan Liittschwager, Stelios Tsampas, Jonathan Castello, Lindsey Kuper

Conference on Algebra and Coalgebra in Computer Science

Higher-Order Mathematical Operational Semantics (Early Ideas)

CALCO 2023

Sergey Goncharov, Stefan Milius, Lutz Schröder, Stelios Tsampas, Henning Urbat

Conference on Algebra and Coalgebra in Computer Science

Talks

Logical Relations in Higher-Order Mathematical Operational Semantics

CHoCoLa

CHoCoLa meeting, Lyon, May 2024 (Invited Talk)

Logical Relations in Higher-Order Mathematical Operational Semantics

WG6 Leuven

WG6 meeting, Leuven, April 2024

A categorical approach to secure compilation

USCD PS

UCSD PS Seminar, February 2021 (Invited Talk)

A summary on categorical contextual reasoning

ACT 2019

Applied Category Theory

Towards Automatic Compartmentalization of C on Capability Machines

FCS 2017

Workshop on Foundations of Computer Security (FCS)

Service

PC Member

- Coalgebraic Methods in Computer Science (CMCS) 2026
- Mathematical Foundations of Programming Semantics (MFPS) 2025
- Principles of Secure Compilation (PriSC) 2023, 2024
- Foundations of Computer Security (FCS) 2022, 2023

Reviewer

- POPL, FoSSaCS, CSL, LICS, CONCUR (various years)

Non-academic Employment

IT Security Engineer (external partner), Anvil Secure

Mar. 2021 – present

IT Security Engineer, Census S.A.

Jun. 2013 – Apr. 2016

Teaching

SDU

Main Instructor

- DM899, *Programming Language Foundations*
Fall 2025
- DM584, *Concurrent Programming*
Spring 2025

KU Leuven

Teaching Assistant

- B-KUL-H01P5A, *Computer Architecture and Software Systems*
Fall 2016, Fall 2017, Fall 2018, Fall 2019, Fall 2020
- B-KUL-H01P1A, *Object-Oriented Programming*
Spring 2017, Spring 2019, Spring 2021
- B-KUL-G0Q40C, *Software Design*
Spring 2018